

Program to find LCM of two numbers

Last Updated : 26 Aug, 2025



Given two positive integers **a** and **b**. Find the Least Common Multiple (LCM) of **a** and **b**.

LCM of two numbers is the smallest number which can be divided by both numbers.

Input : *a = 10, b = 5*

Output : *10*

Explanation : *10 is the smallest number divisible by both 10 and 5*

Input : *a = 5, b = 11*

Output : *55*

Explanation : *55 is the smallest number divisible by both 5 and 11*

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[Naive Approach] Using Conditional Loop

This approach to calculating the Least Common Multiple (LCM) involves starting from the greater of the two numbers and checking if it's divisible by the smaller number. It iterates through multiples of the larger number, incrementing by the larger number itself in each step. The first multiple that is divisible by the smaller number is the LCM. This method is simple and intuitive, but it can be inefficient, especially for large numbers, as it checks multiple values until a match is found.

C++ Java Python C# JavaScript

```
#include <iostream>
#include <algorithm>
using namespace std;

int lcm(int a, int b) {
```



```
// larger value
int g = max(a, b);

// Smaller value
int s = min(a, b);

for (int i = g; i <= a * b; i += g) {
    if (i % s == 0)
        return i;
}

int main() {
    int a = 10, b = 5;
    cout << lcm(a, b);
    return 0;
}
```

Output

10

Time Complexity: $O(\min(a,b))$

Auxiliary Space: $O(1)$

[Expected Approach] Using GCD LCM Formula

*An **efficient solution** is based on the below formula for LCM of two numbers 'a' and 'b'.*

Relation between HCF and LCM

$$\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$$

$$\text{HCF}(16, 28) \times \text{LCM}(16, 28) = 16 \times 28$$

$$\begin{array}{ccccc} 4 & \times & 112 & = & 448 \\ \uparrow & & \uparrow & & \uparrow \\ \text{HCF} & & \text{LCM} & & \text{Product of the Numbers} \end{array}$$

$$a \times b = \text{LCM}(a, b) * \text{GCD}(a, b)$$

$$\text{LCM}(a, b) = (a \times b) / \text{GCD}(a, b)$$

We have discussed [function to find GCD of two numbers](#). Using GCD, we can find LCM.

C++

Java

Python

C#

JavaScript

```
#include <iostream>
using namespace std;

// function for gcd
int gcd(int a, int b) {
    return (b == 0) ? a : gcd(b, a % b);
}

int lcm(int a, int b) {
    return (a / gcd(a, b)) * b;
}

int main() {
    int a = 10, b = 5;
    cout << lcm(a, b);
    return 0;
}
```

Output

10

Time Complexity: $O(\log(\min(a,b)))$

Auxiliary Space: $O(\log(\min(a,b)))$

Java Program to Find LCM of Two Numbers

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